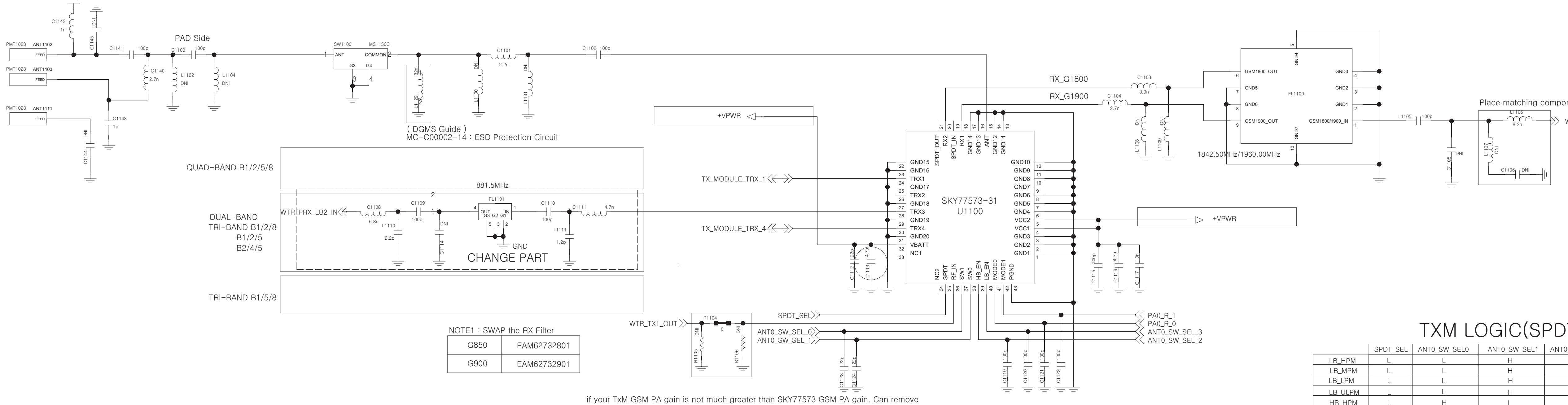
6. CIRCUIT DIAGRAM

< 1-1-1-4_ANT_WCDMA_NO_DRX >

Rev_0.6

ANT1 (GSM Quad / WCDMA B1, B2, B4, B5, B8)



NOTE1 : SWAP the RX Filter

G850	EAM62732801
G900	EAM62732901

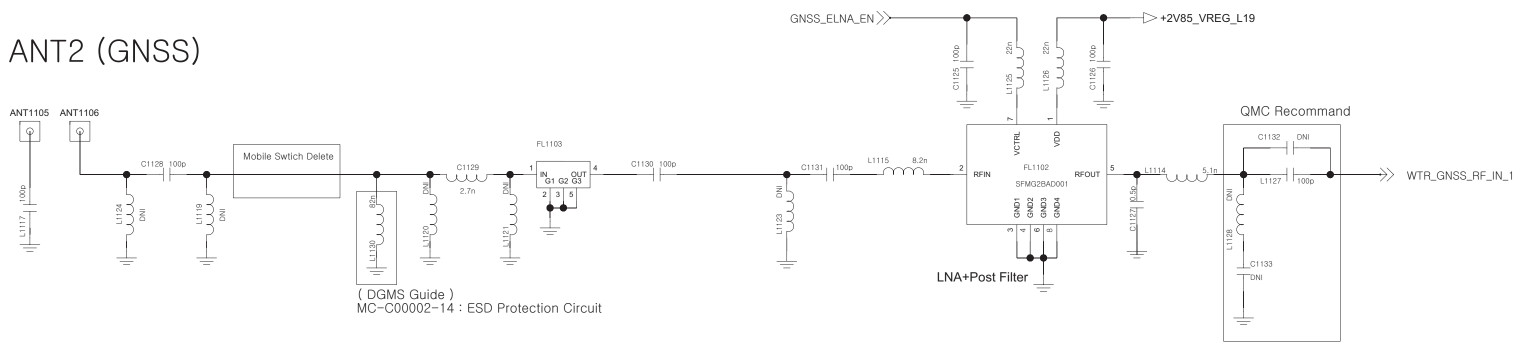
if your TxM GSM PA gain is not much greater than SKY77573 GSM PA gain. Can remove

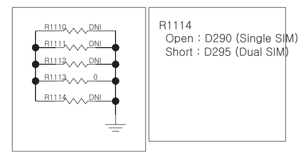
TXM LOGIC(SPDT OFF)

	SPDT_SEL	ANT0_SW_SEL0	ANT0_SW_SEL1	ANT0_SW_SEL2	ANT0_SW_SEL3	PA0_R0	PA0_R1
LB_HPM	L	L	H	L	L	L	L
LB_MPM	L	L	H	L	L	L	H
LB_LPM	L	L	H	L	L	H	L
LB_ULPM	L	L	H	L	L	H	H
HB_HPM	L	H	L	L	L	L	L
HB_MPM	L	H	L	L	L	L	H
HB_LPM	L	H	L	L	L	H	L
HB_ULPM	L	H	L	L	L	H	H
TRX1	H	L	L	L	L	-	-
TRX2	H	L	L	H	L	-	-
TRX3	H	L	H	L	L	-	-
TRX4	H	L	H	H	L	-	-
RX1	H	H	H	H	L	-	-
RX2	H	H	H	H	H	-	-

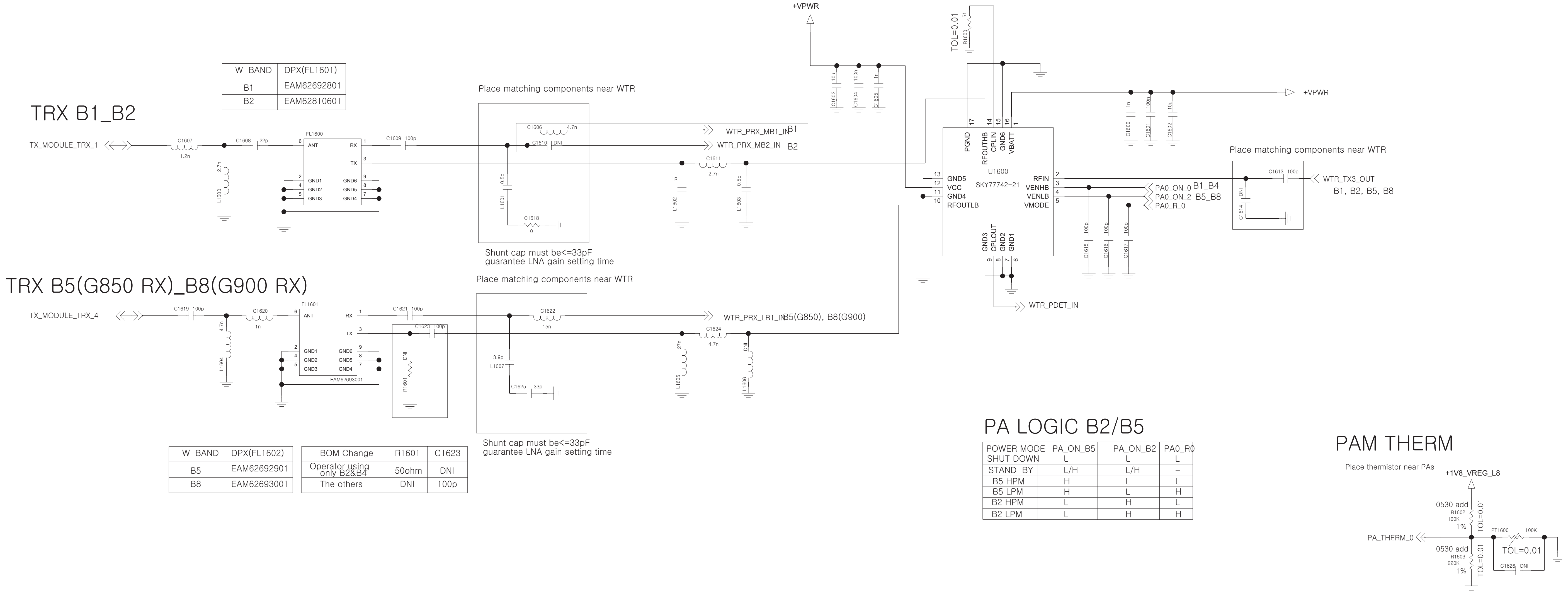
EXT LNA + Filter

ANT2 (GNSS)





< 1-1-4-5 TRX_WCDMA_B1_B2_B5_B8 DUAL > Rev 0.4



W-BAND	DPX(FL1601)
B1	EAM62692801
B2	EAM62810601

W-BAND	DPX(FL1602)
B5	EAM62692901
B8	EAM62693001

BOM Change	R1601	C1623
Operator using only B2&B4	50ohm	DNI
The others	DNI	100p

PA LOGIC B2/B5

POWER MODE	PA_ON_B5	PA_ON_B2	PA0_R0
SHUT DOWN	L	L	L
STAND-BY	L/H	L/H	-
B5 HPM	H	L	L
B5 LPM	H	L	H
B2 HPM	L	H	L
B2 LPM	L	H	H

< 1-1-3-1_RFIC_WTR2605 >

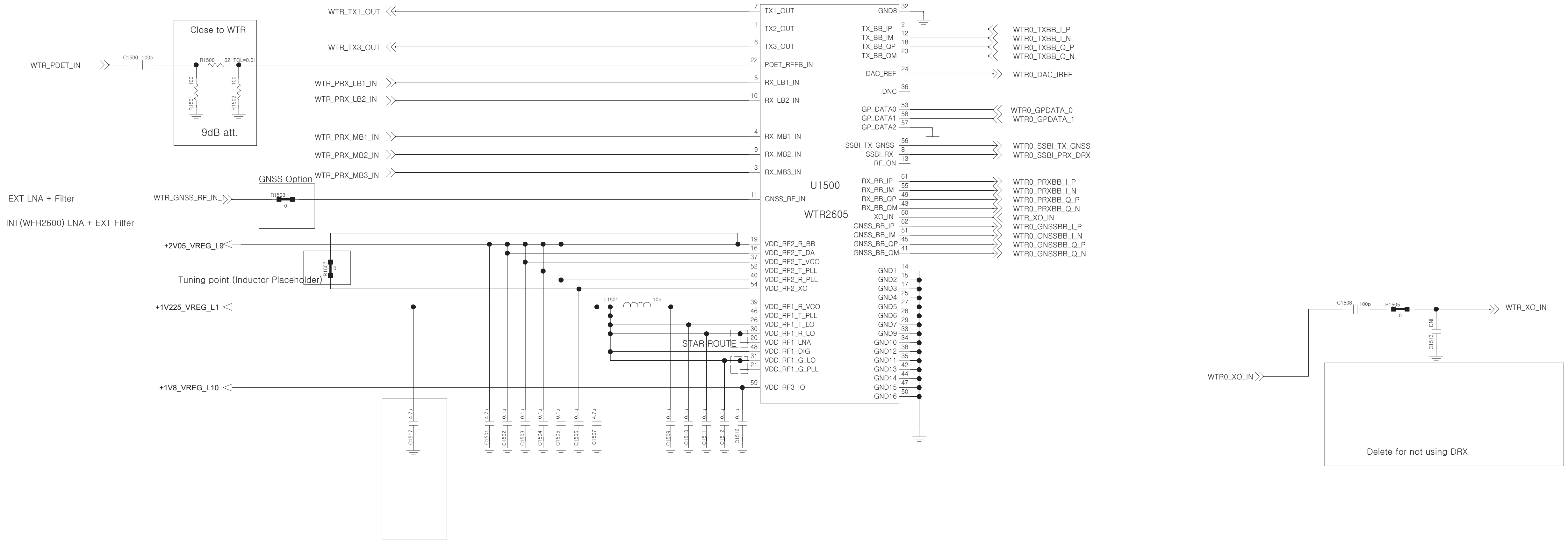
Rev_0.6

CAUTION: YOU MUST CHECK & MODIFY RF PORT AND GPIO ON YOUR SCHEMATICS.

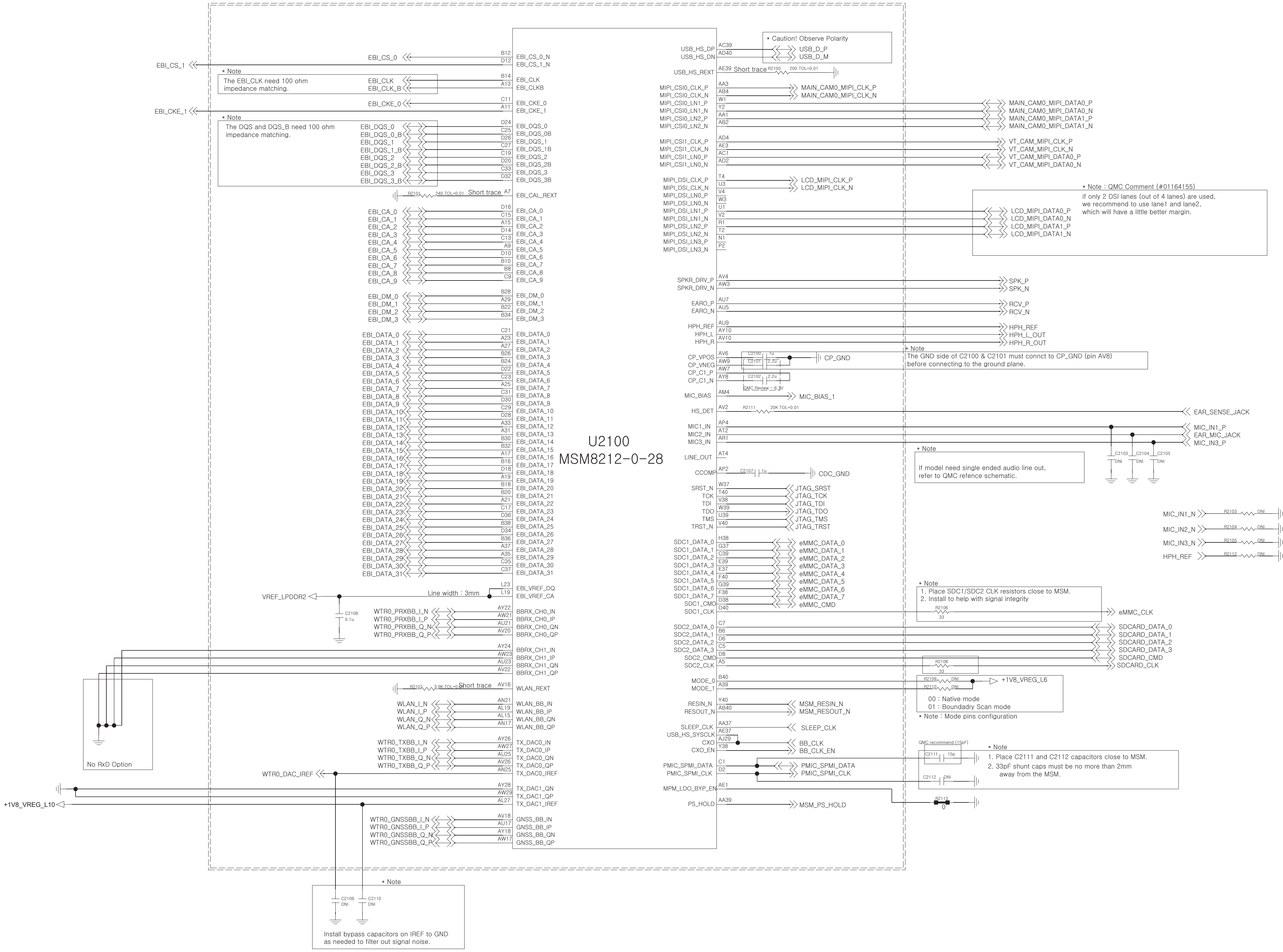
WCDMA							
WTR2605	B2, B4, B5	B1, B2, B5	B1, B2, B8	B1, B5, B8	B2, B5	B1, B5	B1, B8
TX1_OUT	GSM	GSM	GSM	GSM	GSM	GSM	GSM
TX2_OUT	B4	B1	B1	B5			
TX3_OUT	B2, B5	B2, B5	B2, B8	B1, B8	B2, B5	B1, B5	B1, B8
RX_LB1_IN	B5(G850)	B5(G850)	B8(G900)	B8(G900)	B5(G850)	B5(G850)	B8(G900)
RX_LB2_IN	G900	G900	G850	B5(G850)	G900	G900	G850
RX_MB1_IN	B4	B1	B1	B1			B1
RX_MB2_IN	B2	B2	B2		B2	B2	
RX_MB3_IN	G1800,G1900	G1800,G1900	G1800,G1900	G1800,G1900	G1800,G1900	G1800,G1900	G1800,G1900

CDMA			
WTR2605	VZW	SPCS	Metro PCS
TX1_OUT			
TX2_OUT	BC0(BC10)	BC0(BC10)	BC0(BC10),BC15
TX3_OUT	BC1	BC1	BC1
RX_LB1_IN	BC0(BC10)	BC0(BC10)	BC0(BC10)
RX_LB2_IN			
RX_MB1_IN			BC15
RX_MB2_IN	BC1	BC1	BC1
RX_MB3_IN			

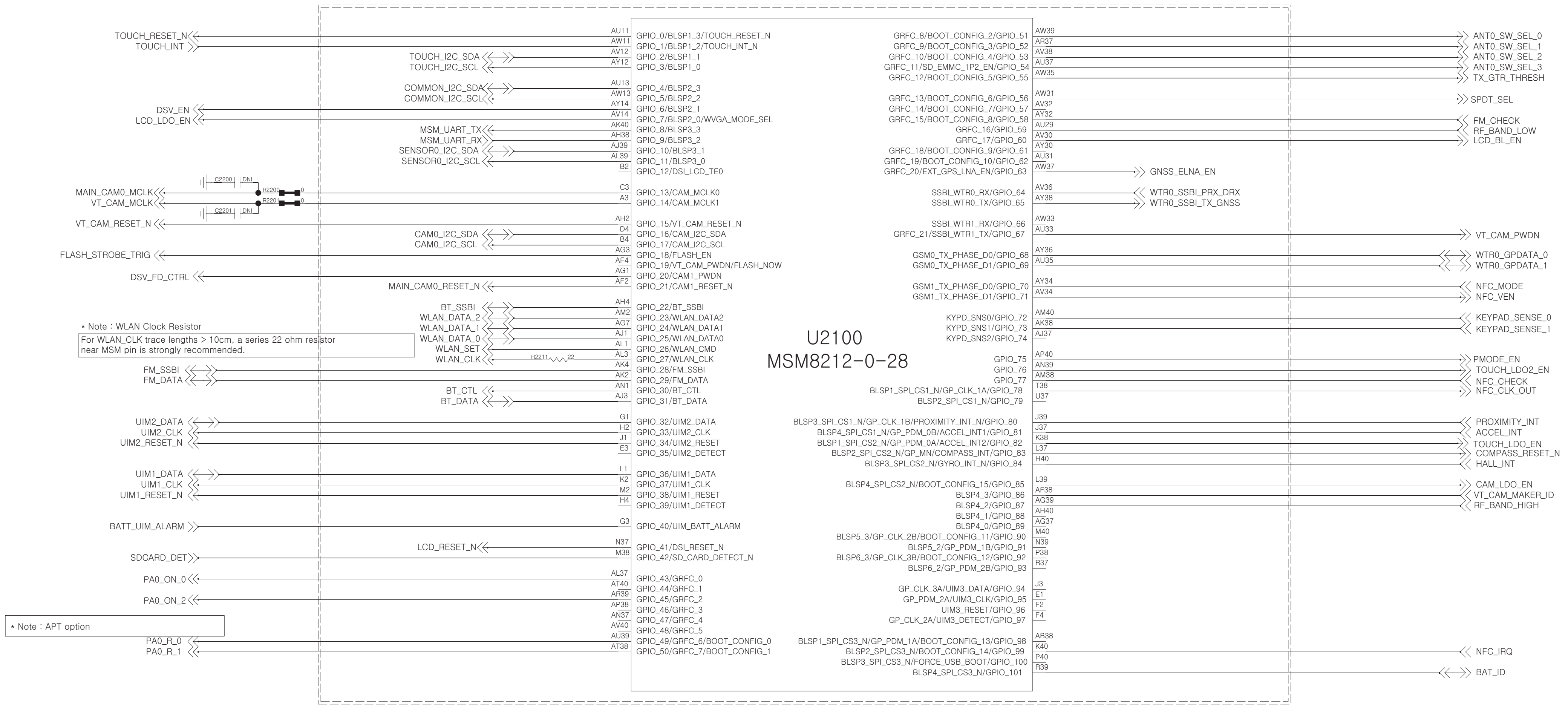
BOM CHANGE	R1606	R1607
WTR_GNSS_RF_IN1	0ohm	DNI
WTR_GNSS_RF_IN2	DNI	0ohm



<2-1-6-1-1_MSM8212-0-28_CONTROL>REV.0.3



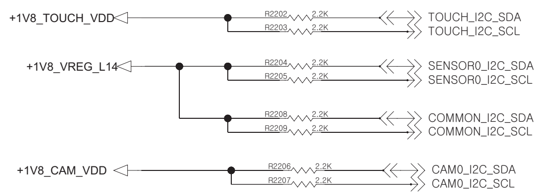
<2-1-6-1-2_MSM8212-0-28_GPIO> REV.0.3



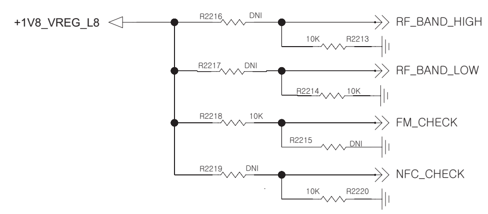
* Note : Secured Boot related GPIOs

1. Do not have pull-ups on the following GPIOs unless intended for boot or secure-boot related configurations : GPIO_61, 62, 85, 90, 92, 98, 99, 100
2. The boot-up problem may happen by external pull-up circuit on these secure-boot related GPIOs

I2C Pull-up

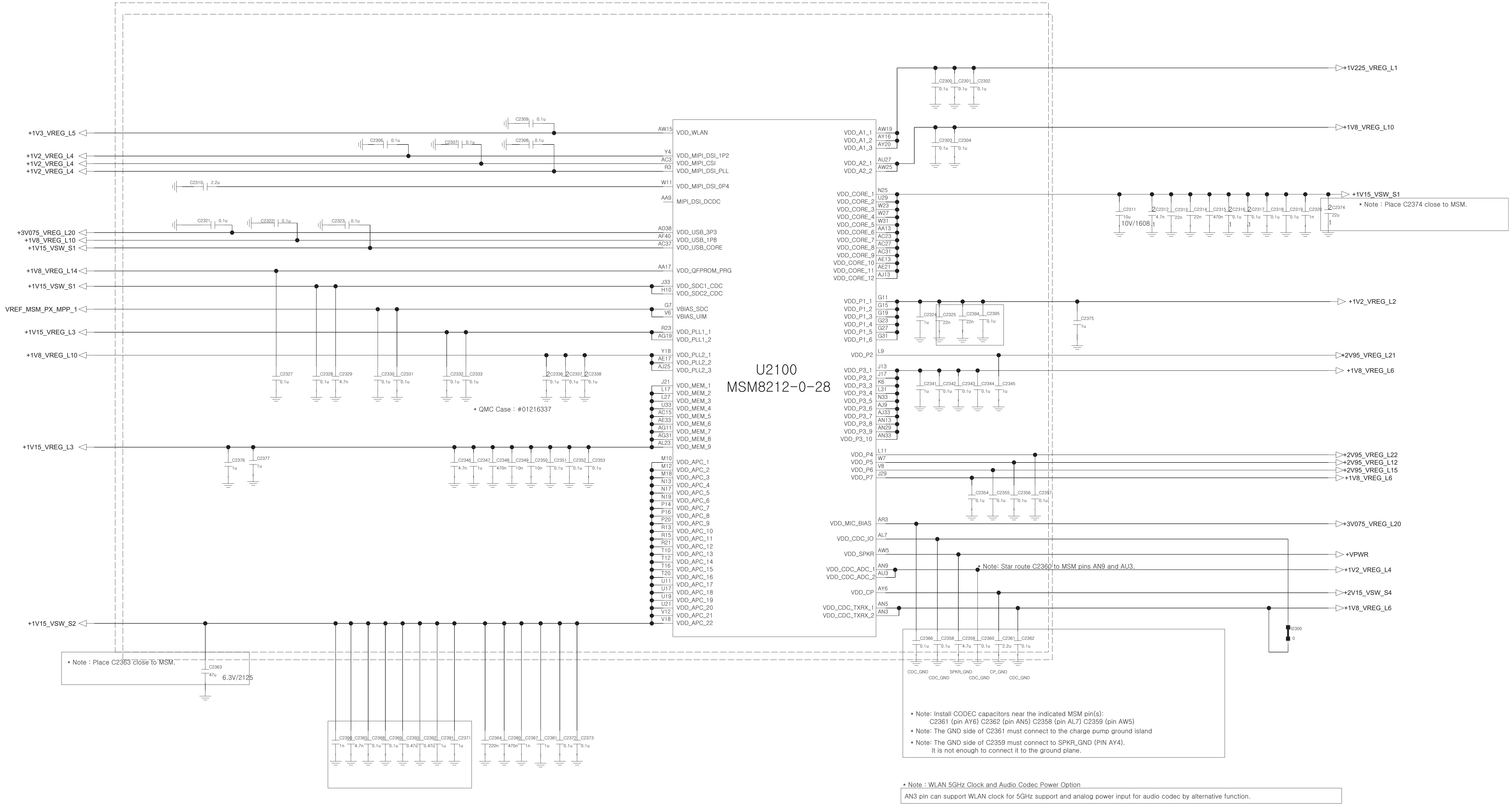


RF BAND Check

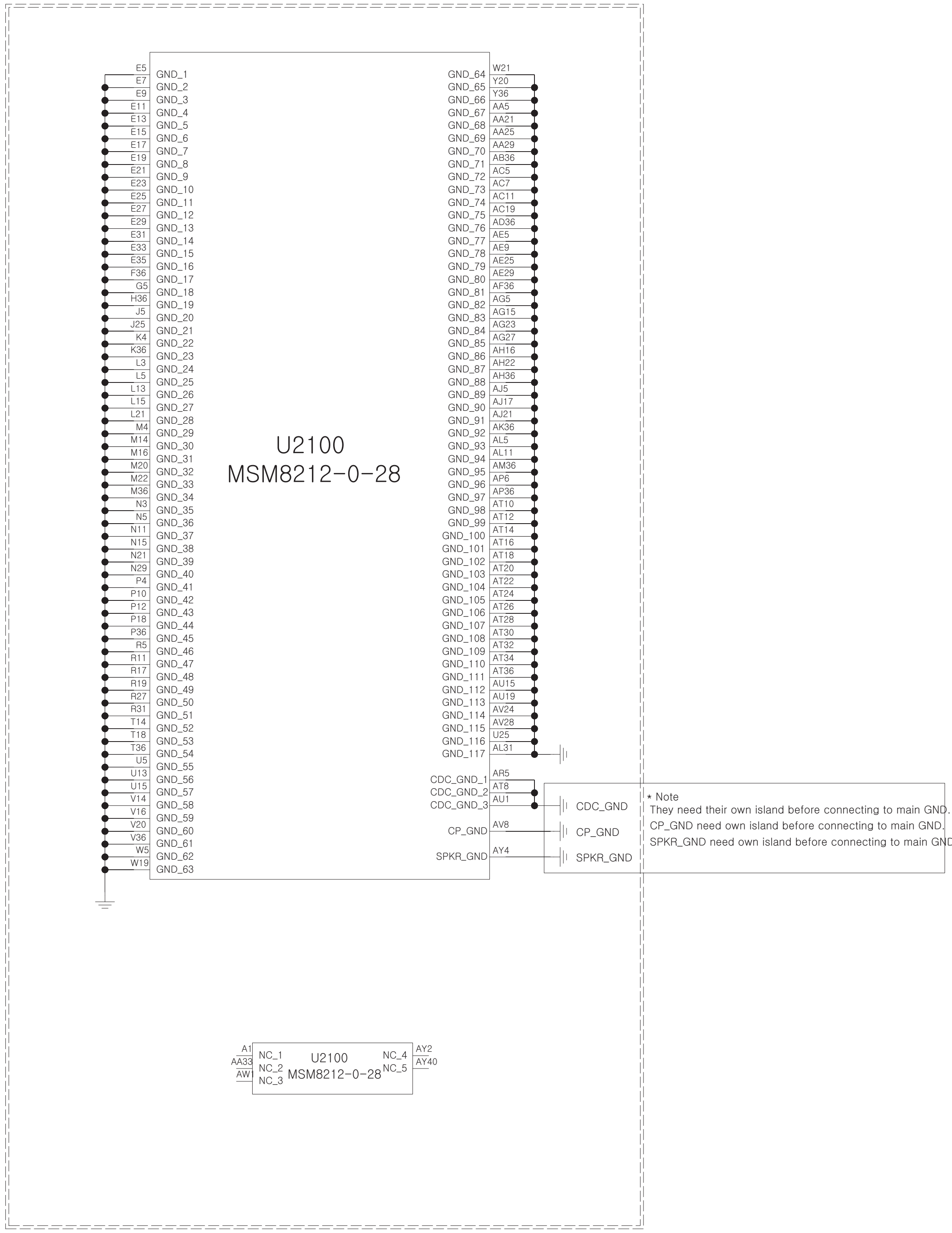


	LOW	HIGH	FM	NFC
GPIO	59	87	58	77
EU_1_8	L	L	L	L
F_1_5	H	L	L	L
G_2_5	H	H	L	L
J_2_8	L	H	L	L
FM	L	L	H	L
NFC	L	L	L	H

<2-1-6-1-3_MSM8212-0-28_POWER> REV.0.3

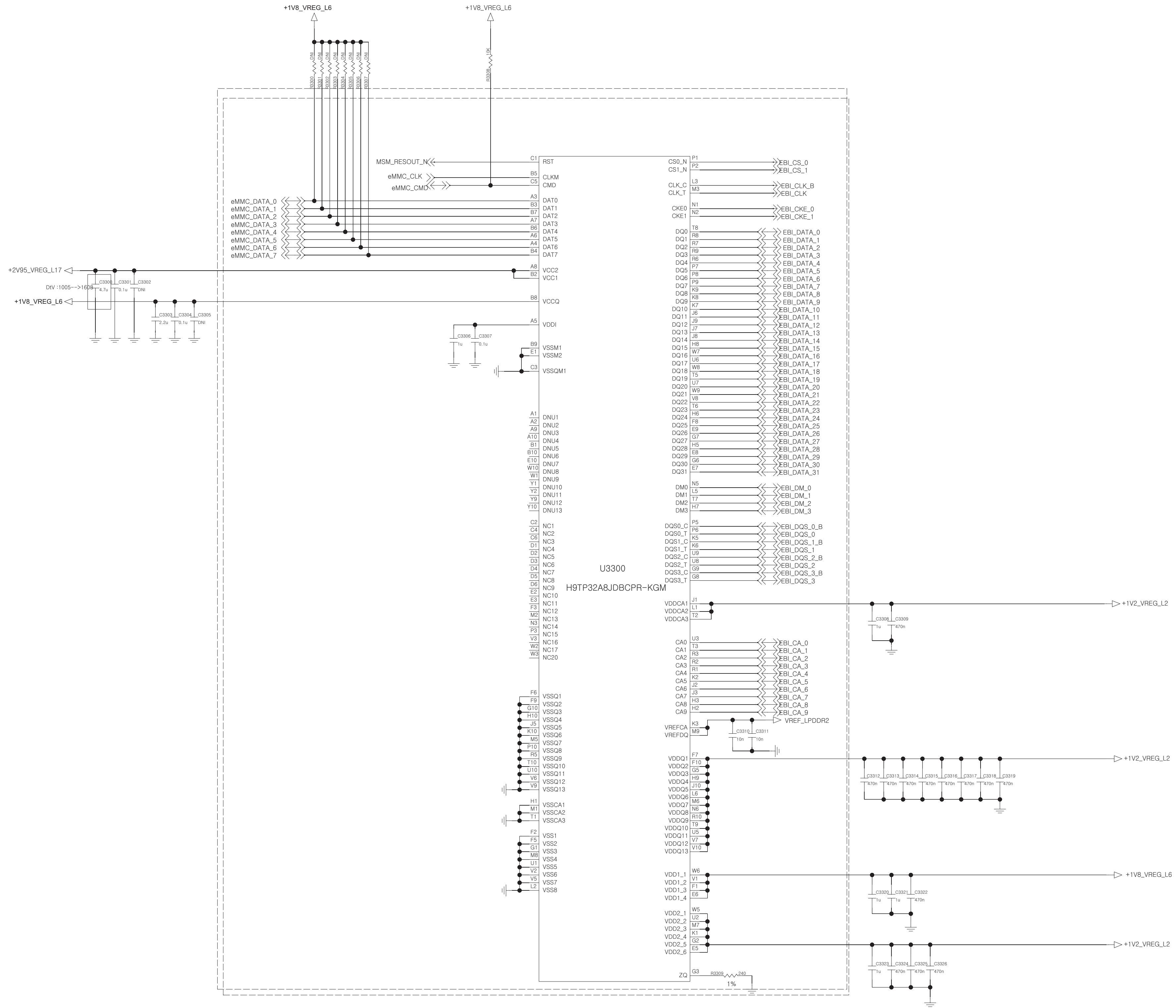


<2-1-6-1-4_MSM8212-0-28_GND/NC>
REV.0.3



< 3_3_2_3_5_MCP_eMMC_v4_5_4G_DDR2_8G_SK_Hynix >

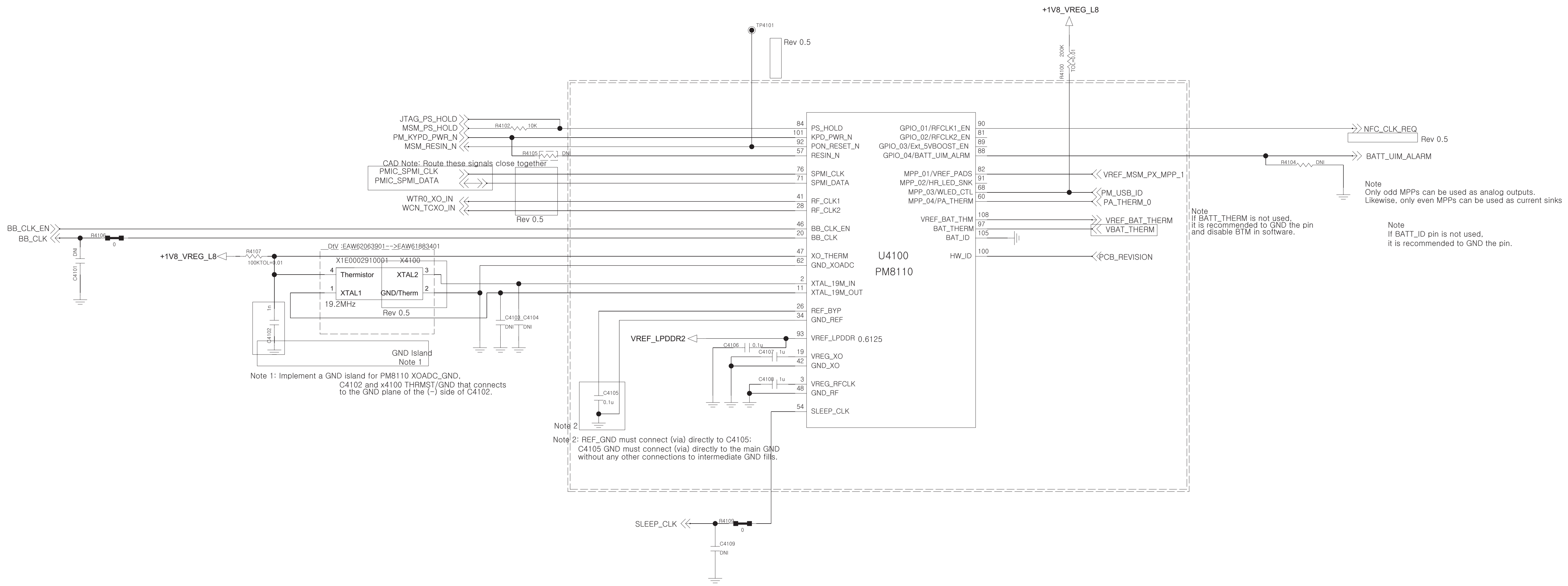
REV_1.0



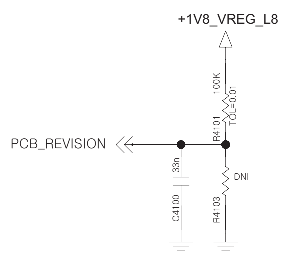
<4-1-4-1_PMIC_PM8110_DATA>

CCDS CARD Information	
Release Date	2013. 12. 2
Based on Reference Schematic	Rev.C
80-ND717-41 MSM8X10 + PM8110 REFERENCE SCHEMATIC	

REV.1.0



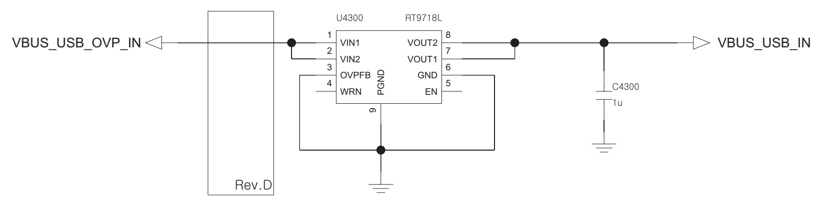
PCB REVISION

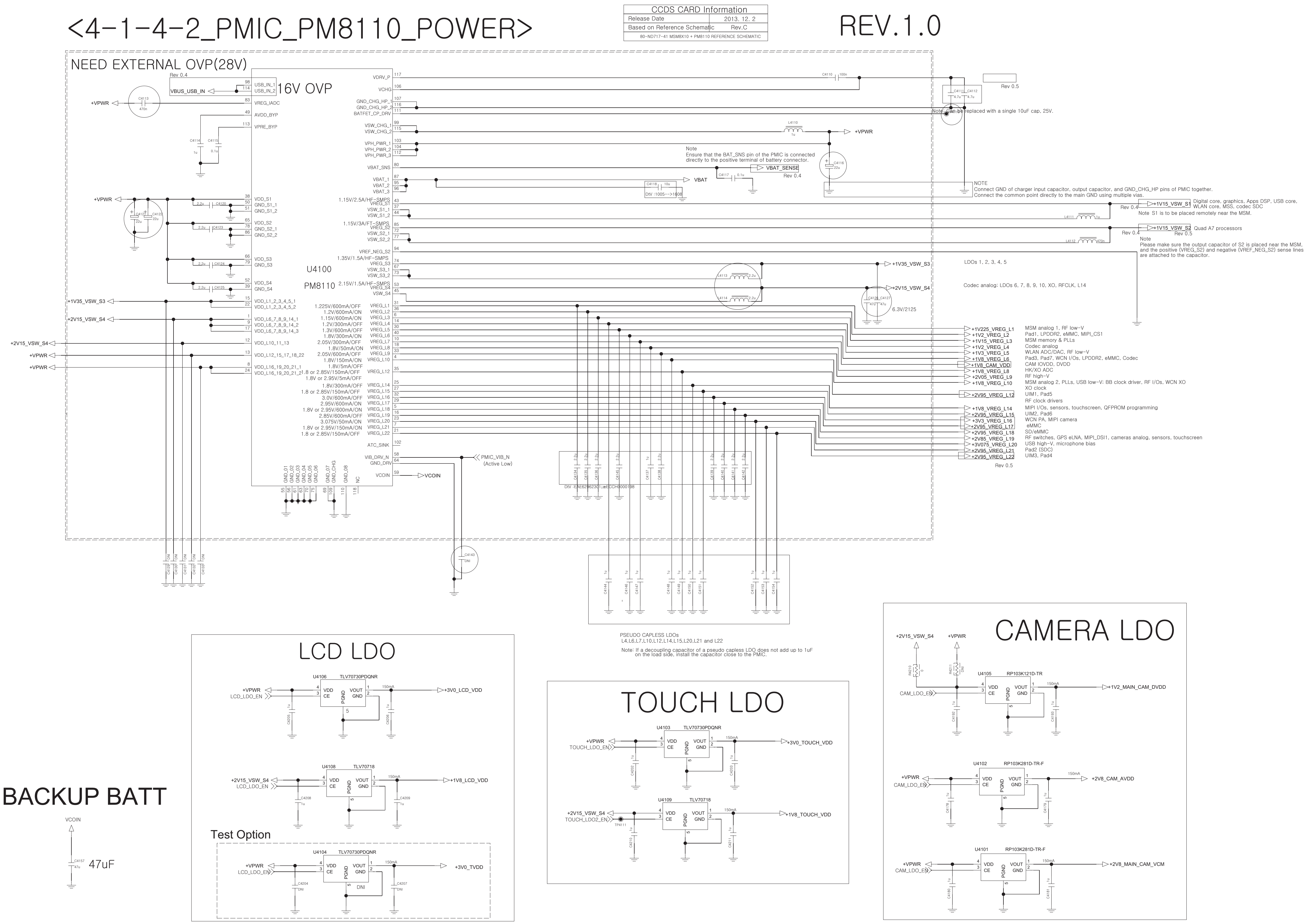


A	100K	20K	0.300
B	100K	27K	0.383
C	100K	39K	0.505
D	100K	51K	0.680
E	100K	75K	0.771
F	100K	100K	0.900
G	100K	130K	1.017
1.0	100K	DNI	1.800
1.1	DNI	100K	0.000

<4-3-3_OVP_RT9718L>

REV.0.3

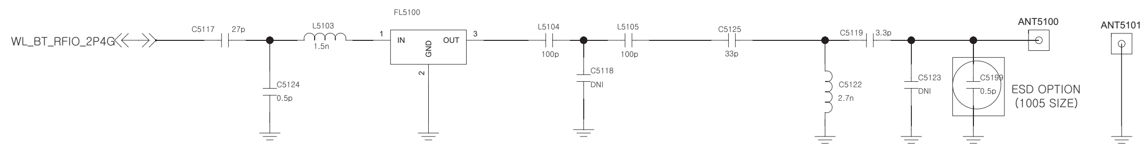



<4-1-4-2_PMIC_PM8110_POWER>
CCDS CARD Information
Release Date 2013. 12. 2
Based on Reference Schematic Rev.C
REV.1.0
NEED EXTERNAL OVP(28V)
16V OVP
U4100
PM8110
LDOs 1, 2, 3, 4, 5
Codec analog: LDOs 6, 7, 8, 9, 10, XO, RFCLK, L14
PSEUDO CAPLESS LDOs
L4,L6,L7,L10,L12,L14,L15,L20,L21 and L22
Note: If a decoupling capacitor of a pseudo capless LDO does not add up to 1uF on the load side, install the capacitor close to the PMIC.
LCD LDO
TOUCH LDO
CAMERA LDO
BACKUP BATT
VCOIN
47uF
Test Option
+3V0_LCD_VDD
+1V8_LCD_VDD
+3V0_TVDD
+3V0_TOUCH_VDD
+1V8_TOUCH_VDD
+1V2_MAIN_CAM_DVDD
+2V8_CAM_AVDD
+2V8_MAIN_CAM_VCM

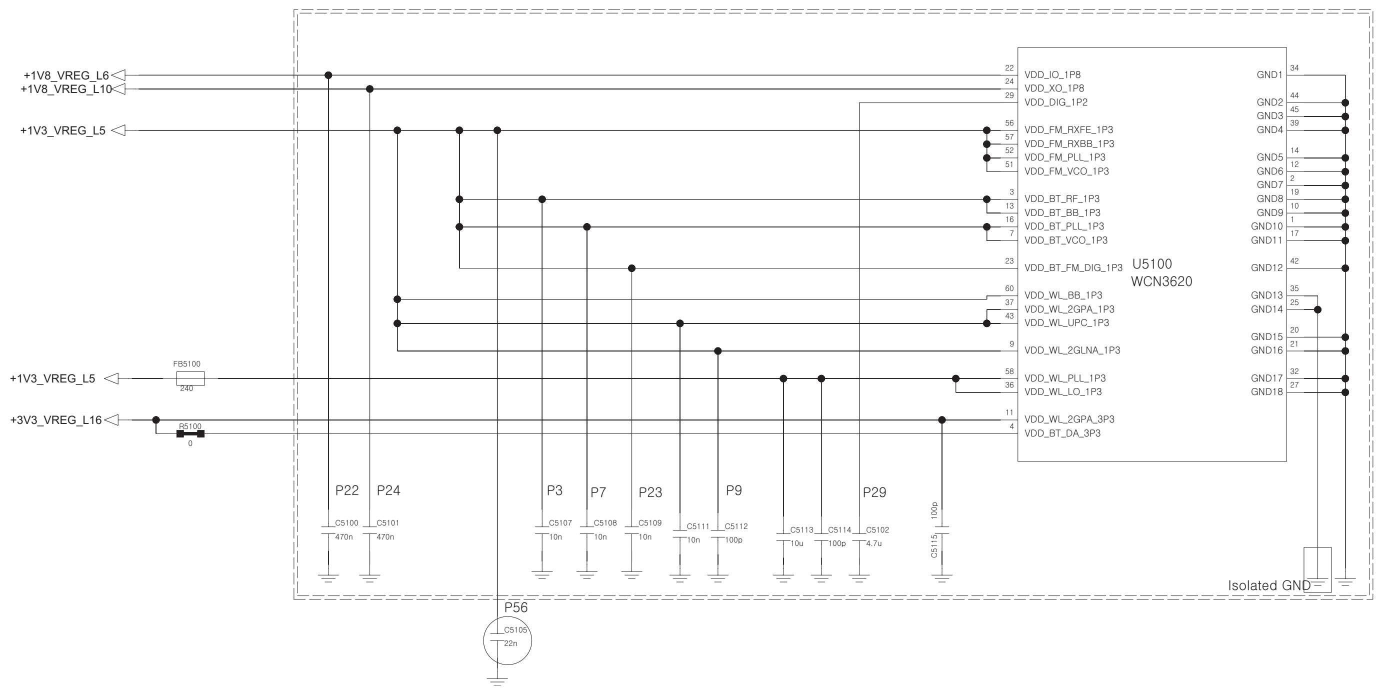
<5-1-1-4> BT_WiFi_Combo_WCN3620

REV 1.0

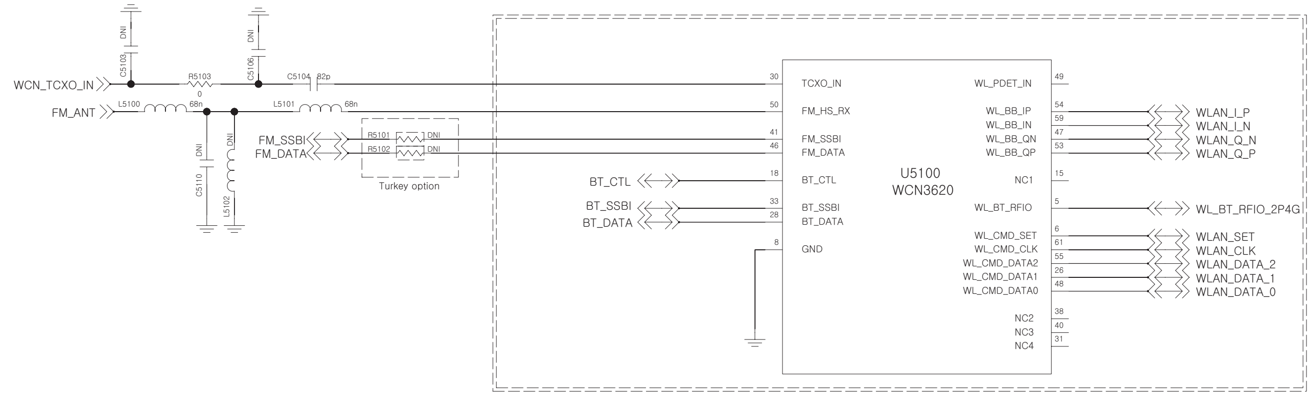
RF Filter



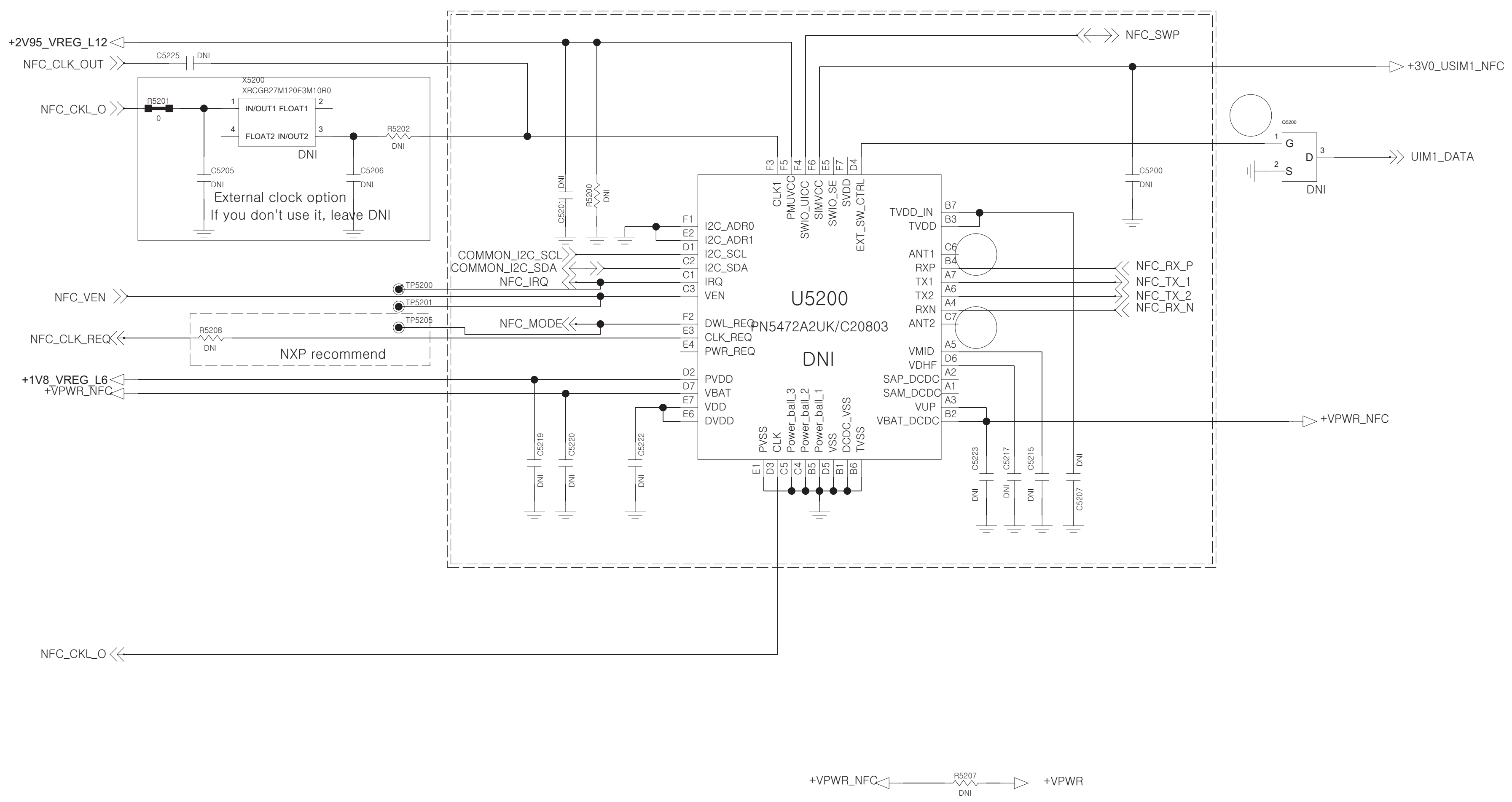
POWER/GND



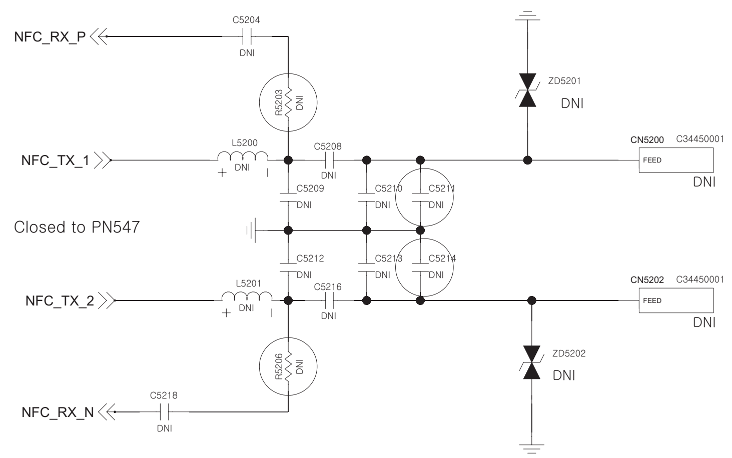
RF/CONTROL



<5-2-1-4_NFC_PN547> Rev_1.1

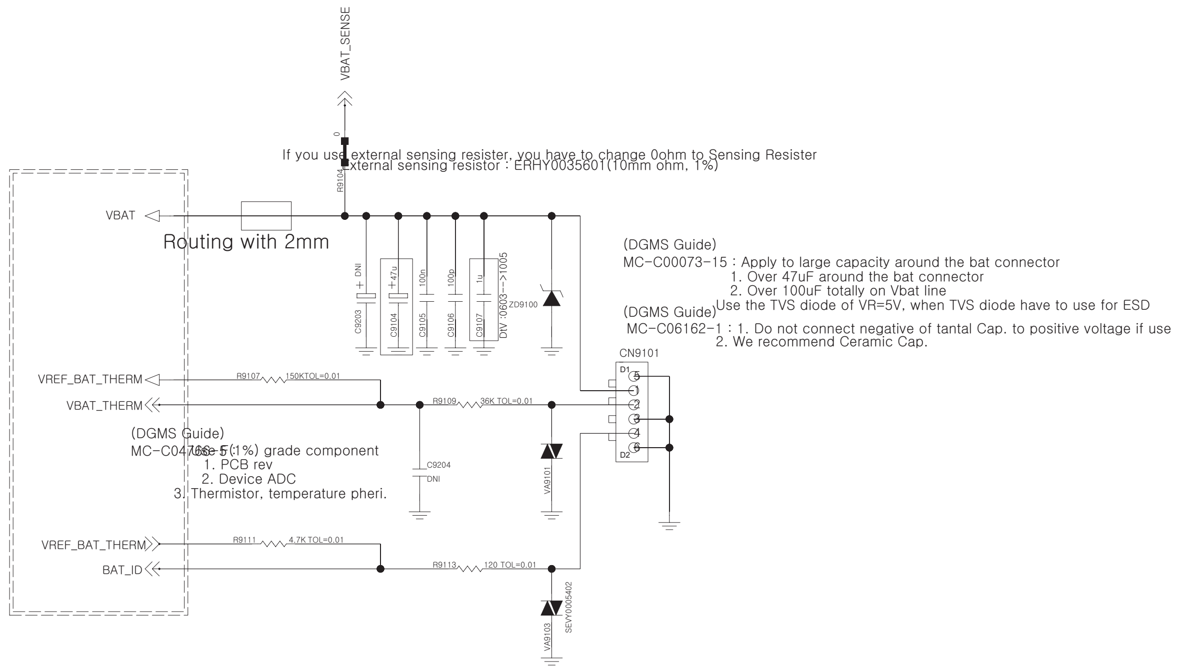


NFC Antenna

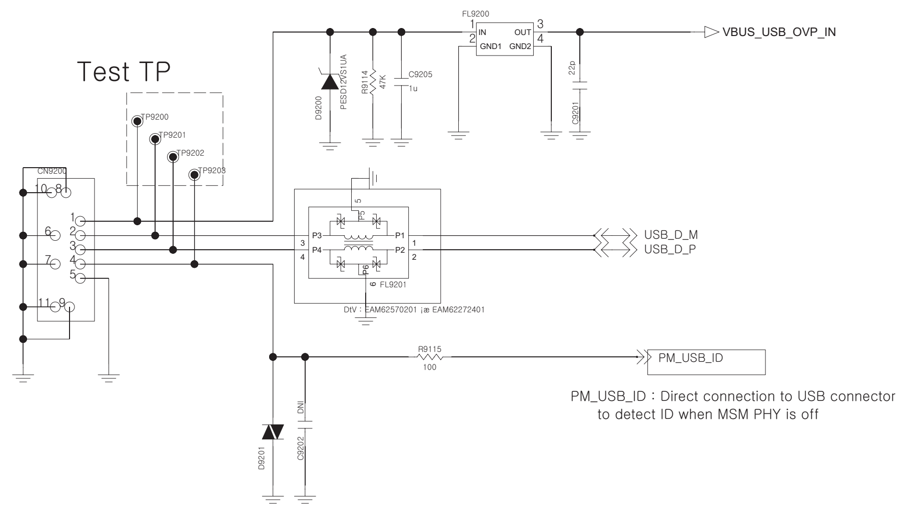


<9-1_Battery_Connector_4Pin>

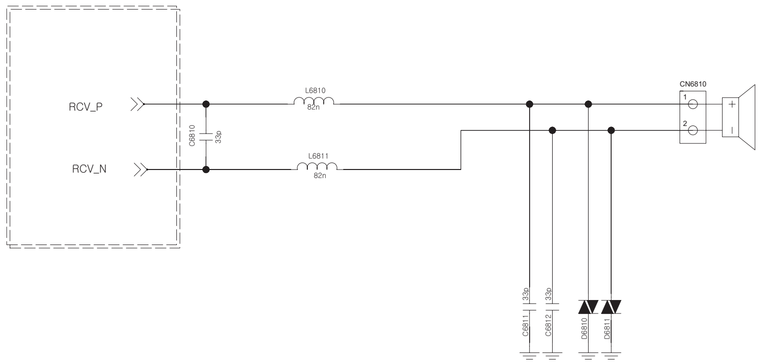
Circuit 1. Batt Conn. /wo EMI Filter



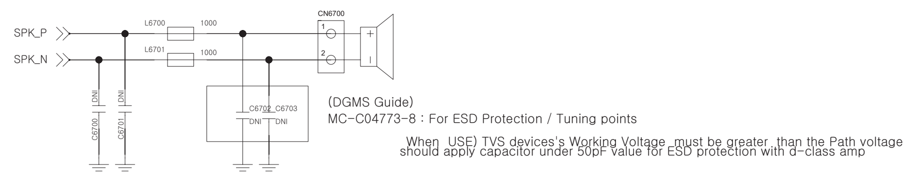
< 9-2-1_IO_USB2.0 > Rev_0.5



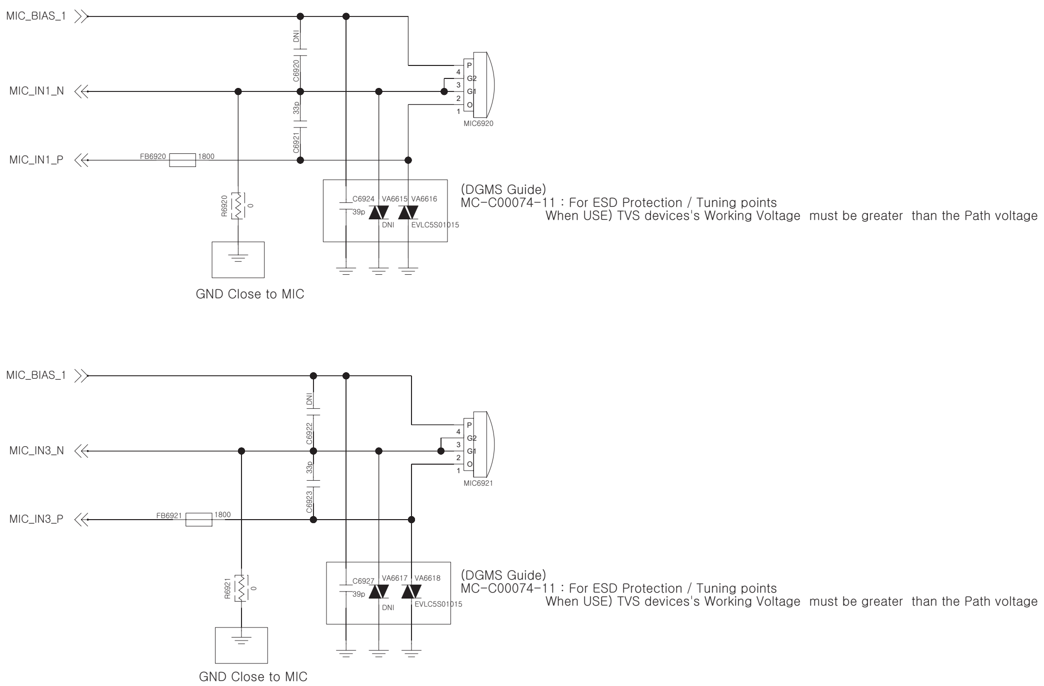
< 6-8-1_Receiver_Only >
Rev_0.3



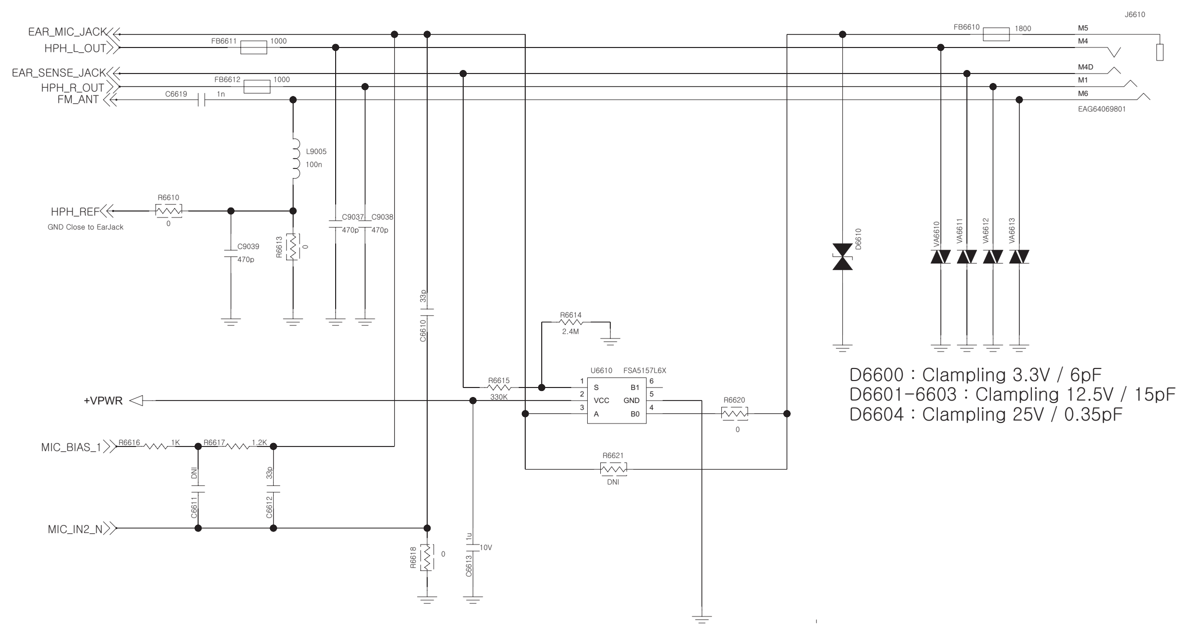
< 6-7-1_Speaker >
Rev_0.4



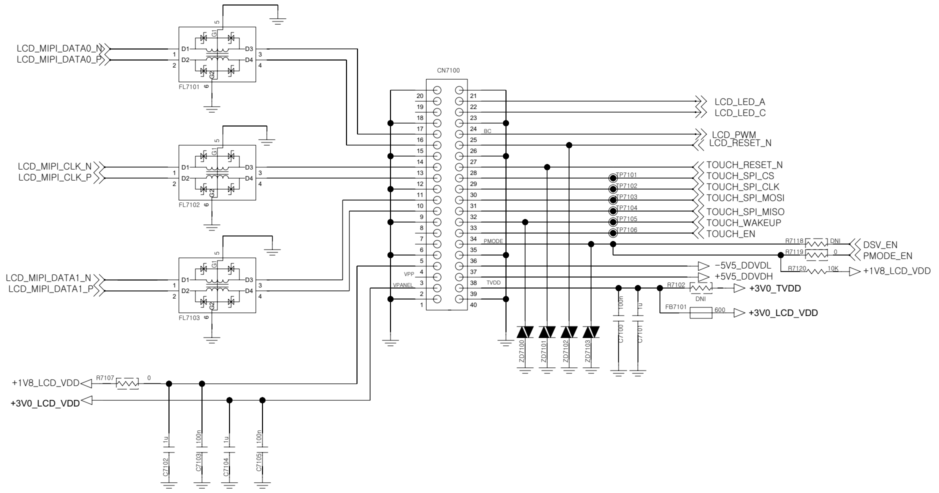
< 6-9-2_2MIC >
Rev_0.4



< 6-6-1_Earjack >



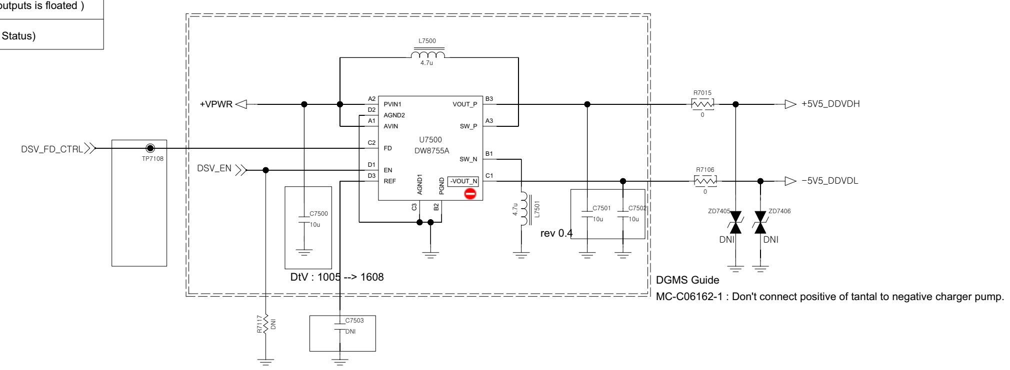
Phase 2 In-cell circuit check!!



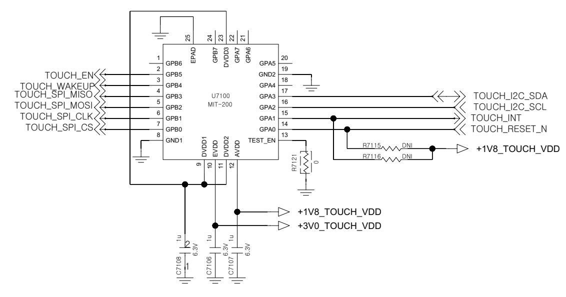
[FD operation Table]

EN	FD	Action
LOW	LOW	Device OFF (Fast discharge active(on) after IC disables)
LOW	High	Device OFF (The status of dual outputs is floated)
High	LOW/High	Device ON (Don't care about FD Status)

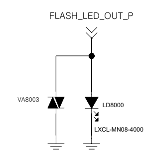
< 7-5-4_DSV_DW8755 >Rev_1.1



Touch MCU

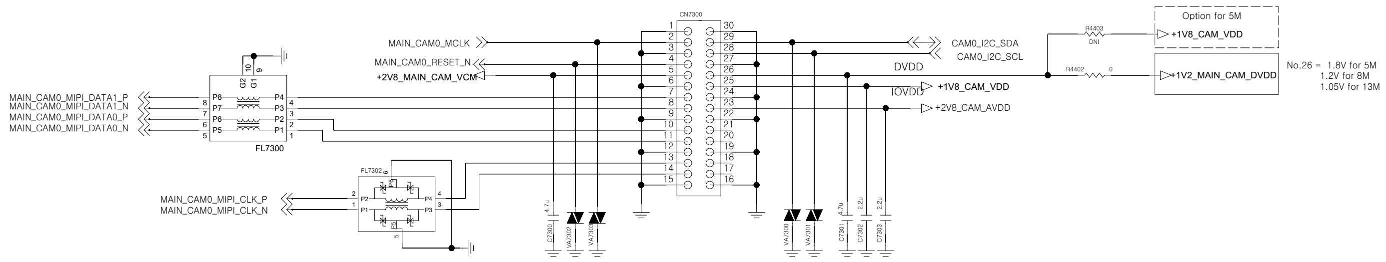


<9-8-3_Flash_LED>

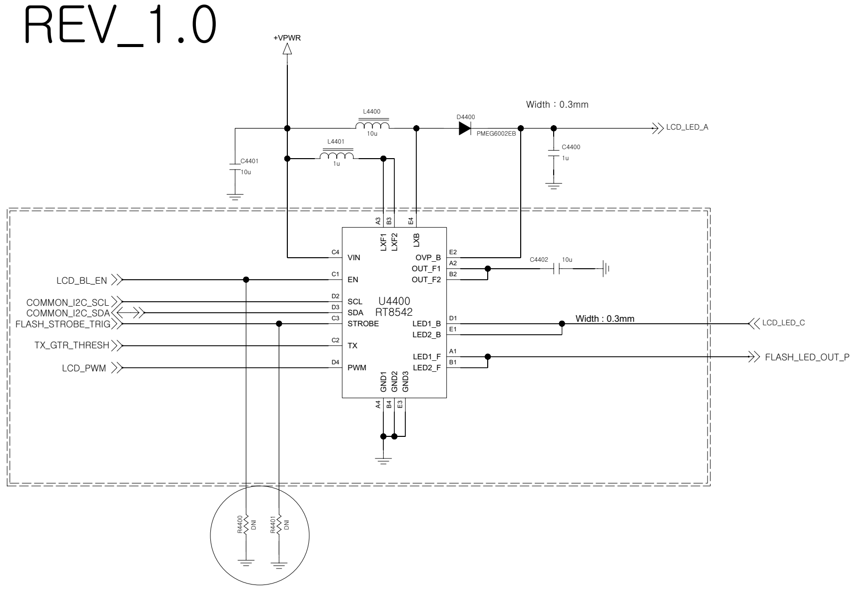


<7-3-5-1_30pin_5M_8M_13M_AF> Rev_0.3

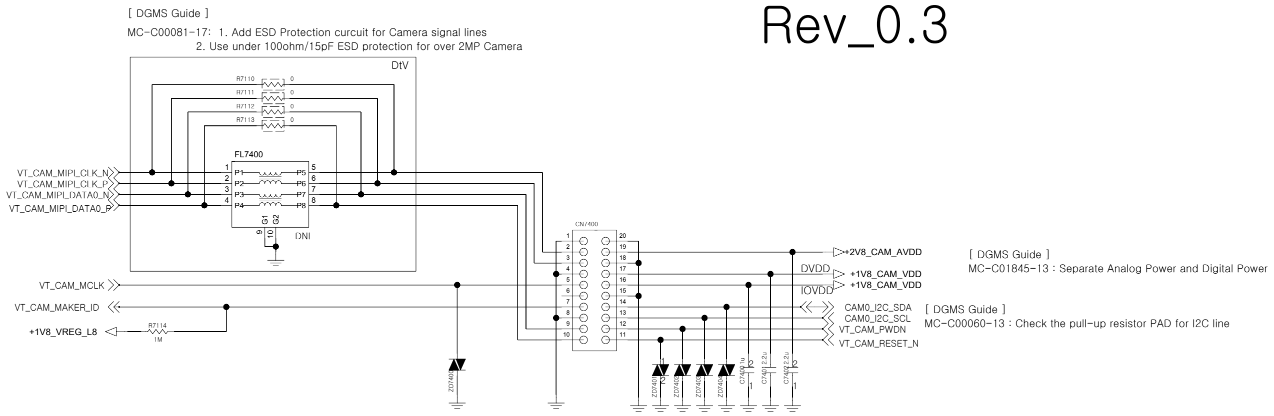
Check the pin map



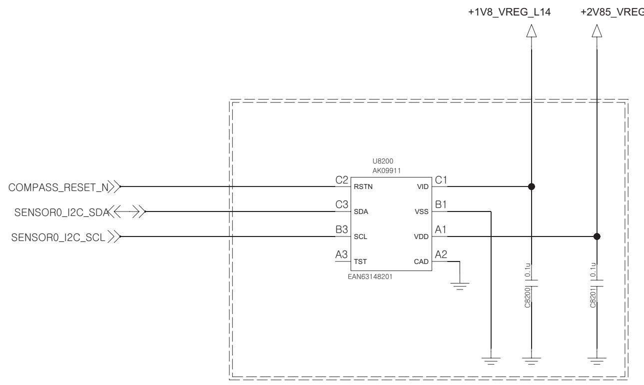
<4-4-3_LCD_Backlight_IC_RT8542> REV_1.0



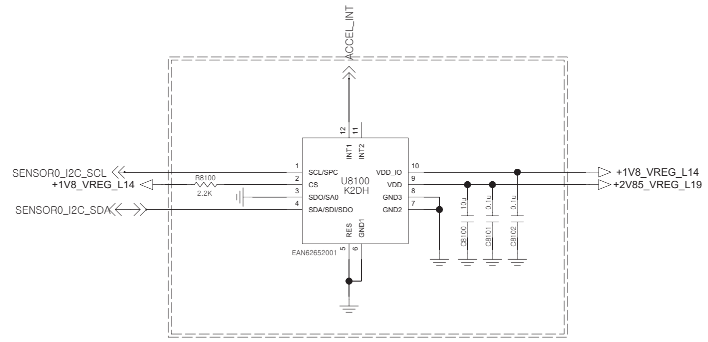
< 7-4-1_VT_Camera_20pin_VGA_1.26M_1.3M_2.4M> Rev_0.3



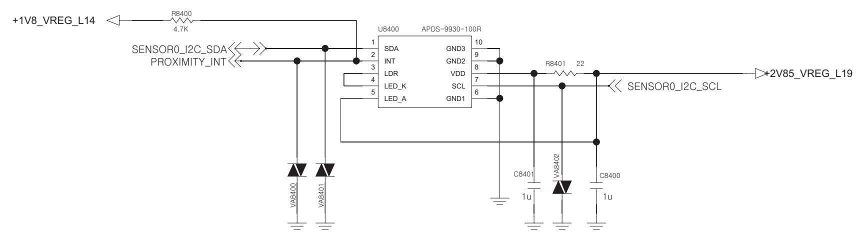
<8-2-1-3_Compass_AK09911 >
Rev_0.3



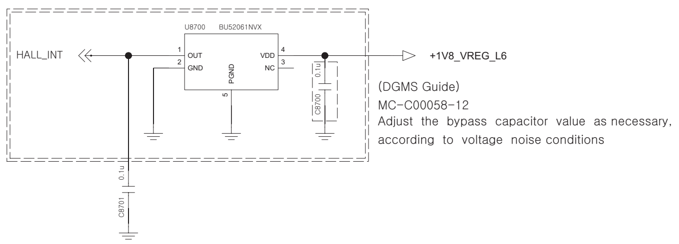
<8-1-1-3_Accelerometer_K2DH >
Rev_0.3



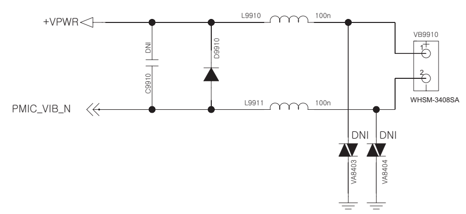
< 8-4-1-2_Proximity_APDS-9190 >
Rev_1.0



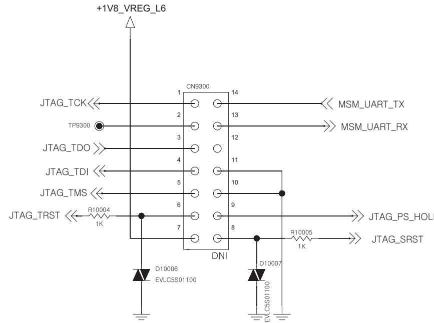
<8-7-1-6_Hall_IC_BU52033NVX>
Rev_1.0



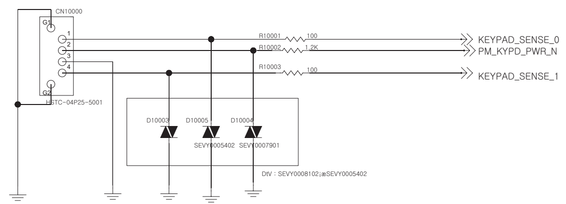
< 9-9-1-1_Motor >
Rev_0.3



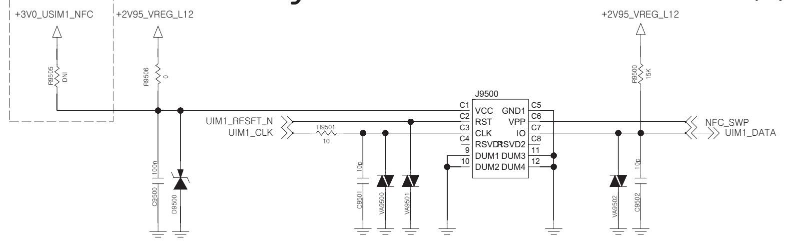
< 9-3-1_JTAG >
Rev_0.4



Back Volume Key

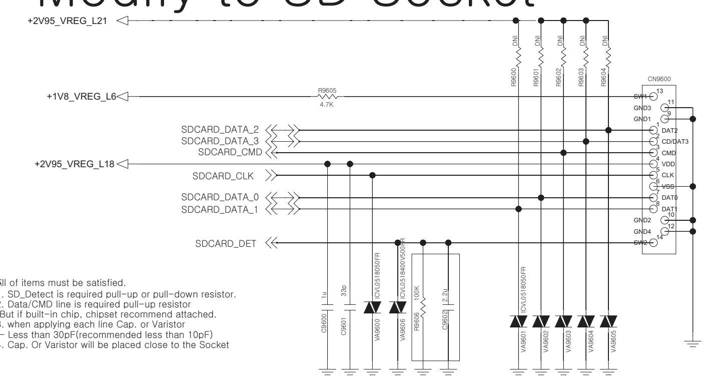


<9-5-2-2_Micro_SIM_8P_1_5T>
Modify to USIM Socket   Rev_0.3



(DGMS Guide)
MC-C04581-13 : All of items must be satisfied.
1. SIM_DATA : It is required pull-up resistor
(But, if built-in chip, chipset recommend attached)
2. when applying each line cap. or Varistor
- Less than 33pF
- Do not use ESD suppressor type
3. when drawing Artwork, SIM_DATA and SIM_CLK line should be separated, and routing.
4. Cap. or Varistor will be placed close to the socket.

< 9-6-1_Socket_Micro_SD >   Rev_0.5
Modify to SD Socket



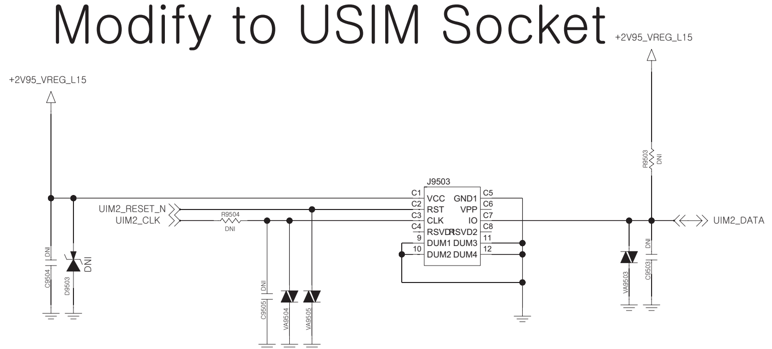
Card Detect	SW1 SW2	SD_CARD_DET
Card Installed		1.8V
Card Not Installed		GND

(DGMS Guide)
MC-C04582-13 All of items must be satisfied.
1. SD_Detect is required pull-up or pull-down resistor.
2. Data/CMD line is required pull-up resistor
(But if built-in chip, chipset recommend attached.
3. when applying each line Cap. or Varistor
- Less than 30pF(recommended less than 10pF)
4. Cap. Or Varistor will be placed close to the Socket

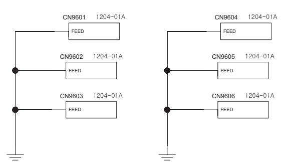
DRBFM Recommendation
Drop / Tumble test Issue.

(DGMS Guide)
MC-C05446-7 All of items must be satisfied.
1. Power(VMMC) of Micr SD card circuit uses independent LDO/PMIC which can be ON/OFF
2. According to Chip type supported by SD card, the current capacity is managed.
: when 200mA High Speed Mode support, the MicroSD Card power is required Power PMIC Port which is obtained margin or LDO
3. SD Card power should not connect the element to serial due to voltage drop.

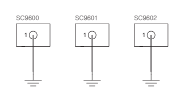
<9-5-2-2_Micro_SIM_8P_1_5T>
Modify to USIM Socket



(DGMS Guide)
MC-C04581-13 : All of items must be satisfied.
1. SIM_DATA : It is required pull-up resistor
(But, if built-in chip, chipset recommend attached)
2. when applying each line cap. or Varistor
- Less than 33pF
- Do not use ESD suppressor type
3. when drawing Artwork, SIM_DATA and SIM_CLK line should be separated, and routing.
4. Cap. or Varistor will be placed close to the socket.



Shield CAN



CAMERA BRACKET